Project Ready: Programming Practice Worksheet

Intro

Ok, so now you should be feeling more confident with finding your way around Studio 5000 Logix Designer and RSLinx Classic.

Programming Problem - Part 1

To increase your confidence with I/O config, tags, aliasing and communication setup complete the following assignment:

I/O Configuration

1. Create a new project in Studio 5000 and call it "Module2_Assignment", make it a 1756-L71 controller and put it in slot 0 of a 7-slot rack.
2. Add I/O to your 7-slot local rack/chassis
    1. Slot 1, add a discrete input module (use DC, 16-point) I prefer 1756-IB16
    2. Slot 2, add a discrete output module (use DC, sourcing 16-point) I prefer 1756-OB16E
    3. Slot 3, add a EtherNet/IP module (1756-EN2T), give it a name
3. Add an EtherNet/IP module to the ethernet network shown in the I/O tree. Make it another 1756-EN2T in a 7-slot rack (remote rack), give it a name
4. Set the IP address to 10.10.10.10 (or anything you like ;))

Add I/O to your 7-slot remote rack/chassis

1. Slot 1, add a discrete input module (use DC, 16-point) choose one you like 🙂
2. Slot 2, add a analog input module (8 channels) choose one you like 🙂

Tags, Aliasing, Addressing

1. Open the Controller Tags folder

2. Create a new tag called "Valve1_is_open" and alias it to the local chassis, slot 1, point 0 (input module)
3. Create a new tag called "Valve1_is_closed" and alias it to the local chassis, slot 1, point 1 (input module)
4. Create a new tag called "Open_Valve1" and alias it to the local chassis, slot 2, point 0 (output module)
5. Create a new tag called "HMI_Start_Request" and make it a BOOL data type.
6. Create a new tag called "Tank_Level" and make it a INT data type.

Programming Problem - Part 2

Now change your controller to the Emulate 5570 controller and download your project to the Logix Emulate controller.

1. Right-click the Controller in the I/O Configuration Tree and choose properties
2. On the General tab, choose “Change Controller”
3. Scroll the list until you find the Emulate 5570 controller

Watch this quick video to change Controller from ControlLogix to Emulate 5570

